ESADE — Perspectives on AI

Fire Watch Ourense

Implementation Plan — Web Dashboard Prototype | April 2026

Project Overview

A Next.js web dashboard for **AI-powered wildfire risk mapping** in the rural areas of Ourense, Galicia (Spain). The system provides real-time fire risk visualization using a **traffic-light system** (Critical High **Medium** Low), integrating weather data (AEMET API), active fire satellite data (NASA FIRMS), and terrain/vegetation analysis.

Target users: Forest agents, local council officers, and emergency responders.

Tech Stack

Category	Technology
Framework	Next.js 16 + React 19 + TypeScript
Styling	Tailwind CSS 4 + clsx + tailwind-merge
Maps	Leaflet 1.9 + React-Leaflet 5
Charts	Recharts 2.15
Icons	Lucide React
Animations	Motion (Framer Motion) 12

1. Shared Utilities & Mock Data

Files: `src/lib/utils.ts` & `src/lib/mock-data.ts`

- `cn()` helper — combines `clsx` + `tailwind-merge`
- **Risk zones:** ~8–10 polygons across Ourense municipalities with risk level, vegetation density, slope grade
- **Fire hotspots:** ~5–6 simulated FIRMS-style points (lat, lng, brightness, confidence, satellite, timestamp)
- **Weather data:** Current conditions (temp, humidity, wind, precipitation, FWI index)
- **Historical trends:** 7-day array of daily risk scores per zone
- **Alerts:** Active warnings (heat wave, high wind, etc.)

2. Interactive Map Component (Leaflet)

File: `src/components/FireMap.tsx` (client component)

- Leaflet map centered on Ourense (~42.34°N, 7.86°W), zoom 10
- **Risk zone polygons** — color-coded by risk level (green → yellow → orange → red)
- **Fire hotspot markers** — pulsing red circles from FIRMS data
- **Popup on click** — zone name, risk score, vegetation type, last updated
- **Layer controls** — toggle risk zones, hotspots, satellite/terrain base maps
- Dynamic import with `ssr: false` (Leaflet requires `window`)

3. Sidebar Panel

File: `src/components/Sidebar.tsx`

3a. Risk Summary (`RiskSummary.tsx`)

- Overall province risk level with traffic-light indicator (large colored circle)
- Count of zones per risk level
- "Last updated" timestamp

3b. Weather Panel (`WeatherPanel.tsx`)

- Current temperature, humidity, wind speed & direction
- Fire Weather Index (FWI) gauge
- Precipitation forecast (next 24h)
- Icons from Lucide

3c. Alerts Panel (`AlertsPanel.tsx`)

- Active warnings list (heat wave, wind advisory, etc.)
- Color-coded severity badges
- Animated entry with Motion

3d. Zone List (`ZoneList.tsx`)

- Scrollable list of all monitored zones
- Each row: zone name, risk badge, trend arrow (up/down/stable)
- Click to zoom map to that zone

4. Analytics Charts (Recharts)

File: `src/components/RiskChart.tsx`

- **7-day risk trend line chart** — one line per zone or aggregated province average
- **Risk distribution bar chart** — zones grouped by risk level
- Responsive container, Tailwind-styled tooltips

5. Main Dashboard Page Layout

File: `src/app/page.tsx`

```
┌──────────────────────────────────────────┐
│  Header: "Fire Watch Ourense" + date + logo  │
```

```
│            │                                     │
├────────────┼─────────────────────────────────────┤
│            │                                     │
│  Sidebar   │          Leaflet Map                │
│  (320px)   │          (flex-1)                   │
│            │                                     │
│  • Risk    │                                     │
│  • Weather │                                     │
│  • Alerts  │                                     │
│  • Zones   │                                     │
├────────────┴─────────────────────────────────────┤
│  Bottom Panel: Risk Trend Charts (Recharts)      │
└──────────────────────────────────────────────────┘
```

- Full viewport height (`h-screen`)
- Responsive: sidebar collapses to drawer on mobile
- Dark theme optimized for control-room use

6. Styling & Theming

Design tokens (dark theme):

Element	Color
Background	`#0f172a` (dark navy)
Cards	`#1e293b` (slate)
Low risk	`#22c55e` Green
Medium risk	`#eab308` **Yellow**
High risk	`#f97316` Orange
Critical risk	`#ef4444` Red

7. Machine Learning Model — Fire Risk Prediction

7.1 Algorithm: Random Forest Classifier

We use a **Random Forest** model for fire susceptibility mapping, validated by the reference paper *(Piao et al., 2022)* which achieved an **AUC of 0.835** on a comparable task using Google Earth Engine. Random Forest handles the non-linear relationships between vegetation density, terrain slope, and thermal data better than simpler regression models.

Output: A probability score (0–1) per grid cell, mapped to a traffic-light classification for user adoption:

Probability	Risk Level	Color	Action
0.0 – 0.25	Low	Green	Normal monitoring
0.25 – 0.50	Medium	Yellow	Increased vigilance
0.50 – 0.75	High	Orange	Preventive brush-cutting, patrol priority
0.75 – 1.0	Critical	Red	Emergency alert, resource pre-positioning

7.2 Input Features (Data Layer Dictionary)

Data Source	Variable	Purpose	Temporal Resolution
Sentinel-2 (Multispectral)	NDVI / Moisture index	Biomass density & vegetation health — primary fuel indicator	5 days (revisit cycle)
Sentinel-3 / MODIS	LST (Land Surface Temperature)	Heat accumulation — real-time thermal anomaly detection (pre-ignition)	Daily
DEM — SRTM	Slope / Aspect	Topography — fire spread speed (fires accelerate uphill)	Static (one-time)
AEMET API	Wind speed, humidity, precipitation	Weather forcing — hyperlocal dynamic conditions	Hourly (real-time)

Key improvement over reference paper: We add real-time thermal input (Sentinel-3/MODIS LST) and live weather data (AEMET), which the reference paper lacked. This transforms the model from *static long-term susceptibility* to *dynamic daily risk prediction* — a critical requirement from our user interviews.

7.3 Training Pipeline

Platform: Google Earth Engine (GEE) — cloud processing, no heavy local infrastructure needed.

1. **Define study area:** Ourense province boundary polygon in GEE (swap from reference paper's Gangwon-do region)
2. **Collect training labels:**
    - **Positive samples (fire = 1):** Historical fire perimeters from EFFIS (European Forest Fire Information System) and NASA FIRMS hotspot archive for Ourense (2015–2025)
    - **Negative samples (fire = 0):** Randomly sampled points from non-burned areas, balanced 1:1 ratio with positive samples
3. **Extract features per sample point:**
    - NDVI from Sentinel-2 composites (median of fire season: June–September)
    - Slope and aspect from SRTM DEM (30m resolution)
    - LST from MODIS/Sentinel-3 (mean of fire season)
    - Distance to roads, distance to urban areas (auxiliary spatial features)
4. **Train/test split:** 70% training / 30% testing, stratified by fire/no-fire class
5. **Train Random Forest in GEE:**
    - `ee.Classifier.smileRandomForest(numberOfTrees: 100)`
    - Variable importance ranking to validate which features drive predictions
6. **Evaluate:** AUC-ROC, confusion matrix, overall accuracy. Target: AUC ≥ 0.80
7. **Generate prediction map:** Classify every pixel in Ourense → export as GeoTIFF risk layer

7.4 From Static to Dynamic (Real-Time Layer)

The base Random Forest model produces a **structural susceptibility map** (which areas are inherently fire-prone). To deliver **daily dynamic risk**, we overlay real-time modifiers:

Modifier	Source	Effect on Risk Score
Current temperature > 35°C	AEMET API	+0.15 to base score
Humidity < 30%	AEMET API	+0.10 to base score
Wind speed > 30 km/h	AEMET API	+0.10 to base score
No rain in last 7 days	AEMET API	+0.10 to base score
Active thermal anomaly nearby	FIRMS / MODIS LST	+0.20 to base score

Final dynamic risk = clamp(base_score + modifiers, 0, 1)

This approach gives users the "where will the danger be *this afternoon*" predictions that forest agent Rafael explicitly requested.

7.5 Model Update Cycle

- **Base model retrain:** Annually (after each fire season, add new fire perimeters to training set)
- **NDVI / vegetation layer:** Updated every 5 days (Sentinel-2 revisit)
- **Weather modifiers:** Updated hourly (AEMET API)
- **Thermal anomalies:** Updated every 6 hours (FIRMS near-real-time)

7.6 Validation Strategy

- **Quantitative:** AUC-ROC ≥ 0.80, precision/recall per risk class, spatial cross-validation (leave-one-municipality-out)
- **Qualitative:** Validation with stakeholders — show predicted risk maps to Rafael (forest agent) and Maria (council officer) and compare against their field knowledge of high-risk zones
- **Temporal holdout:** Train on 2015–2023 fires, test on 2024–2025 fires to confirm temporal generalization

8. API Integration Points (Future)

API	Purpose	Env Variable
AEMET	Real-time weather data for Ourense	`AEMET_API_KEY`
NASA FIRMS	Active fire satellite detections	`FIRMS_MAP_KEY`

For the prototype, all data is mocked. API routes can be added at `src/app/api/weather/route.ts` and `src/app/api/fires/route.ts` .

9. Final File Structure

```
src/
├── app/
│   ├── layout.tsx          # Root layout, Geist fonts, metadata
│   ├── page.tsx            # Main dashboard page
│   ├── globals.css         # Dark theme, Leaflet overrides
│   └── favicon.ico
├── components/
│   ├── FireMap.tsx          # Leaflet map with risk zones
│   ├── Sidebar.tsx          # Sidebar container
│   ├── RiskSummary.tsx      # Traffic-light risk overview
│   ├── WeatherPanel.tsx     # Weather conditions panel
│   ├── AlertsPanel.tsx      # Active warnings
│   ├── ZoneList.tsx         # Monitored zones list
│   └── RiskChart.tsx        # Recharts trend charts
└── lib/
    ├── utils.ts             # cn() helper
    └── mock-data.ts         # Prototype mock datasets
```

10. Build & Run

```
npm run dev      # Dev server → http://localhost:4000
npm run build    # Production build
npm run start    # Production server → http://localhost:4000
```

Fire Watch Ourense — Perspectives on AI & Sustainability — ESADE 2026
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